CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc Biotechnology/ Microbiology/ Biochemistry Examination May 2018

Course code & Name: MS762 Immunology

03.05.2018 Thursday 01:30 PM To 02:00 PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **20**

1. BCG is a _________ vaccine

(i) monovalent (ii) trivalent
(iii) pentavalent (iv) heptavalent

2. PAMP stands for

(i) Pathogen-associated molecular print (ii) Pathogen-associated molecular pattern
(iii) Pathogen-associated molecular plan (iv) All of the above

3. Which of the following in NOT a blood group type

(i) ABO (ii) Rh
(iii) MNSs (iv) Xx

4. A person with "hh" genotype represent

(i) Presence of Bombay and ABO blood group (ii) Presence of Bombay and absence ABO blood group
(iii) Absence of Bombay and ABO blood group (iv) Absence of Bombay and presence ABO blood group

5. NK cells kill their target cell by

(i) lysing the cell
(ii) inducing inflammation
(iii) inducing apoptosis
(iv) all of the above

6. Immune system comprises of

(i) Two anti-parallel systems
(ii) Three anti-parallel systems
(iii) Two parallel interrelated systems
(iv) Three parallel systems

7. Select the right statement about MHC molecules

(i) They are secreted exclusively by B-cells
(ii) There are four classes of them
(iii) They recognize PAMPS
(iv) They are recognized by T-cells

8. Antibodies are secreted by

(i) Pro B-cells
(ii) Hematopoietic stem cells
(iii) Plasma cells
(iv) All somatic cells except RBCs

9. Select the right series of events seen during a humoral response

(i) Interleukins or helper T cells costimulate B cells> Antigens bind to B cells> B cells proliferate and produce plasma cells>The plasma cells secret antibodies>Released antibodies circulate through the body and binding to antigens

(ii) B cells proliferate and produce plasma cells> The plasma cells secret antibodies>Interleukins or helper T cells costimulate B cells> Antigens bind to B cells> Released antibodies circulate through the body and binding to antigens

(iii) Antigens bind to B cells>Interleukins or helper T cells costimulate B cells> B cells proliferate and produce plasma cells>The plasma cells secret antibodies>Released antibodies circulate through the body and binding to antigens

(iv) All the above series of events are wrong

10. Complement system consist of a group of

(i) Proteins
(ii) proteins and carbohydrates
(iii) Lipids
(iv) Lipids and Nucleic acids

11. Select the right combination

(i) Endocytic pathway/ Endogenous antigens/ MHC Class II
(iv) Endocytic pathway/ Exogenous antigens/ MHC Class I
(iii) Cytosolic pathway/ Exogenous antigens/ MHC Class II
(ii) Cytosolic pathway/ Endogenous antigens/ MHC Class I

12. End result of all the complement pathways is formation of

(i) TAC
(ii) MAC
(iii) NAC
(iv) SAC

13. DTH is type ______ hypersensitivity

(i) I
(ii) II
(iii) III
(iv) IV

14. SCID is associated with

(i) Transplantation immunology (ii) Immunodeficiency

(iii) Autoimmunity (iv) Hypersensitivity

15. Select the correct inflammatory event

(i) Pain at the affected body part (ii) Redness at the affected body part

(iii) Body part's function is lost (iv) All the above events occurs

16. Expand mAbs ____________

17. Expand TLR ______________________

18. Innate immunity is also known as specific immunity

True False

19. The complement system is a part of the both innate and adaptive immunity

True False

20. Cancer cells can be killed by immune cells

True False

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II Semester of MSc Biotechnology/ Microbiology/ Biochemistry Examination May 2018

Course code & Name: MS762 Immunology

03.05.2018 Thursday 02:01 PM To 04:30 PM Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **20**

1. Compare innate and adaptive immunity **5**

OR

Describe briefly, T-cell activation and differentiation

2. Describe **ANY ONE** **5**

(i) Antigen processing and presentation pathway for endogenous antigens
(ii) How diversity in antibodies' light chains is achieved?

3. How mAbs are generated using hybridoma technology **5**

OR

Describe alternative pathway of complement activation

4. Attempt **ANY TWO** **5**

(i) An antibody is a dimer of dimers. Justify
(ii) Discuss Bombay blood group
(iii) Compare antigenicity and immunogenicity

SECTION – II

Q-III Answer the following questions as directed **30**

1. What is "an abnormal immune response against self-components" called? How it is generated and types of diseases it leads to? **5**

OR

Briefly describe different hypersensitivity types

2. What do you understand by HAMA? Explain how it can be avoided? **5**

OR

Derive Scatchard equation and explain its importance

3. Draw schematic diagram of MHC molecules **5**

OR

What do you understand by "graft" in transplantation immunology? Mention its various types and explain the role of T cells in allograft rejection

4. How AlphaLISA differs from ELISA? Describe AlphaLISA in detail **5**

OR

Describe pregnancy test using mAbs

5. Briefly describe various method of generating currently available vaccines **5**

OR

How T_H and T_C cells can be separated using flowcytometric method

6. Attempt **ANY TWO** **5**

(i) Discuss the role of Peter Medawar in transplantation immunology
(ii) What is DNA vaccine?
(iii) How Chickens avoid anthrax infection?